

Report on the “Computer Vision in Archaeology” Training School in Brno, Czech Republic (14-18 September 2026)

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Attending the Training School on “Computer Vision in Archaeology” at the Institute of Archaeology at the Czech Academy of Sciences, proved to be a very positive experience. Though time constraints almost dissuaded me from attending, I ultimately decided to participate because the themes and concepts covered appeared to align well with my own research. And quite the correct hunch that was! The wide range of topics covered helped me to both to broaden and to deepen my understanding of Computer Vision applications and their potential integration into cultural heritage management and archaeology workflows. Indeed, I leave this school with a better grasp on fundamentals of the field, which I can already envision helping me throughout my future studies, both theoretically and practically.

Day 1

The first day included gentle introductions to basic concepts of Computer Vision for archaeology, agentic programming, as well as, more broadly, programming with the Python language. The opening presentation contextualized the applicability of Computer Vision for handling specialized tasks in our field, underscoring both problems it could solve, but also plausible obstacles and caveats we might face or need to address. Working with CLIP for zero-shot classification helped me better understand the functionality of pertinent algorithms, in addition to giving me the chance experiment with my own datasets, thus also aiding me in better understanding their scope (Fig. 1). Throughout this first hour, I also found the discussion rounds to be quite fruitful; they gave us the opportunity to get a basic idea of what other attendees had experienced thus far, and how they had thought to overcome conceptual or practical issues they might have faced. Additionally, I thought that the analysis of the statistical report produced by the pre-school survey was more than interesting, in that it allowed a first quantification of everyone’s experience with and understanding of the topics at hand.

Then, the overview of agentic modeling introduced to me a tool that I had been entirely unfamiliar with -or, admittedly, slightly intimidated by. This was only the first of many practical exercises I am sure to return to! Finally, the practical exercise for programming with Python was additionally useful. Despite my previous experience with the language, it was helpful to come back to some basic topics, and to also acquire access to a practical guide for baseline coding.

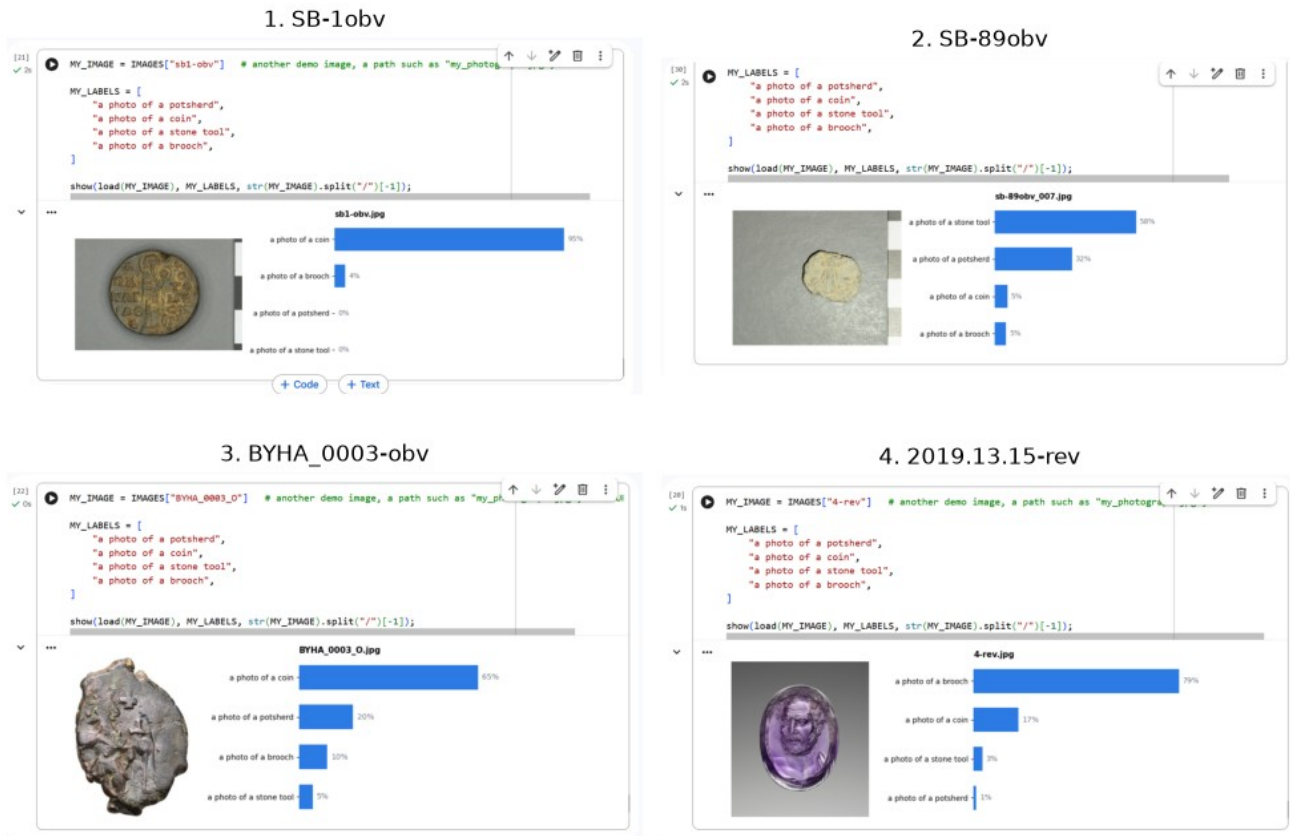


Figure 1: Figure 1. Selection of tested image from my own datasets (1, 2: Robert Feind Collection/Byzantine-era lead seals. 3: Numismatic Collection of the University of Cologne/Coin. 4: Getty Museum Collection/Brooch

Day 2

The second day was dedicated to dataset annotation with CVAT; yet again, another tool I had had no previous encounters with. Working practically with realistic archaeological datasets gave me the opportunity to consider how such a workflow may be applied to tasks relevant to my research going forward. In fact, being given the chance to upload my own images meant that I could, yet again, enjoy experimenting with pertinent data. This presentation helped me understand exactly why data annotation can be such a cumbersome task, but also why it's an important preparatory step for any Computer Vision project, which needs to be deliberated upon intensely, in order to ensure a that good foundation is set for the former. The day concluded with presentations from the participants (including myself). All projects presented were extremely memorable. The diversity of datasets and approaches was especially positive, as it provided an inspiring look into practical applications of what we had been taught previously, or what was coming ahead.

Days 3 and 4

The presentation on the third day concerned microscopic imagery and the difficulties in handling such complex datasets. Then, on the fourth day we explored Computer Vision for inspecting and studying satellite images and Scandinavian rock art. All three projects showcased diverse methods for analyzing specialized heritage data, which I had not previously considered. What stood out to me most from those, were the creative solutions to handling what had appeared to be otherwise insurmountable tasks.

Indeed, when considering object detection for any application, I had thought that attempting to fine-tune an algorithm for in-the-wild implementations, or even to re-train a machine learning model based on the such fine-tuned algorithms was more than challenging. While the presenters made it clear that careful thought and planning had to go into developing the workflows they presented, seeing the results of their hard work and discussing their implementation- and thought-processes helped make everything appear considerably more approachable. Once again, I am sure to return to the relevant Notebooks, in order to study them more thoroughly and try to replicate them practically, with my own images and datasets. Because, despite the seeming discrepancies between the ones cited and the ones I work with, similar concepts might be interesting regardless of their exact contents.

During these days we were also offered tours of the facilities of the Institute of Archaeology. Needless to say, I was impressed by the equipment and archive thereof!

Day 5

The final day was devoted to a dataset of coins and covered various approaches for analyzing that. Due to the nature of the dataset and the methods presented, this application was clearly more in-line with what I had done previously. Therefore, I found it extremely useful for my future research. The integration of monocular depth estimation for a task I had not previously considered was fascinating and I am excited to try out something similar myself. The notions towards 3D Computer Vision were something that I had mentioned might be interesting to bring up when filling out the aforementioned pre-school survey, but I did not expect to actually encounter anything entirely so relevant. This day allowed me to get in touch with more people that had experience with what I had studied intensely for my Master Thesis, and I look forward to exchanging information, data, resources, and ideas with them going forward.

Social Events

Another highlight of this training school were the social events sprinkled throughout the program. The various activities planned for us permitted us to both discover the city and the Institute of Archaeology, but also to come closer together in a meaningful way. The entire team of organizers and attendees was welcoming and approachable, making it easy to come into contact with everyone and to create what I hope will be long-lasting connections. In fact, I would go as far as to say that this was the most successful such event I have attended in terms of team chemistry, since I truly enjoyed spending time with anyone and everyone, both during the school, but also afterwards. The city of Brno offered an excellent backdrop for all this, since it emerged as a vibrant and lively city we did not get bored of exploring. Here, too, I learned something new; Czech beer is, indeed, the best in Europe, but Czech wines can be a bit too sweet (though that is not always a negative).

Conclusion

The training school in Brno, Czechia, was undoubtedly a memorable experience, both in terms of academic development, but also social interactions. The insights gained and connections made will follow me throughout my academic career. I applaud the organizers and presenters for the great event they hosted and I cannot wait to see what this team does next!